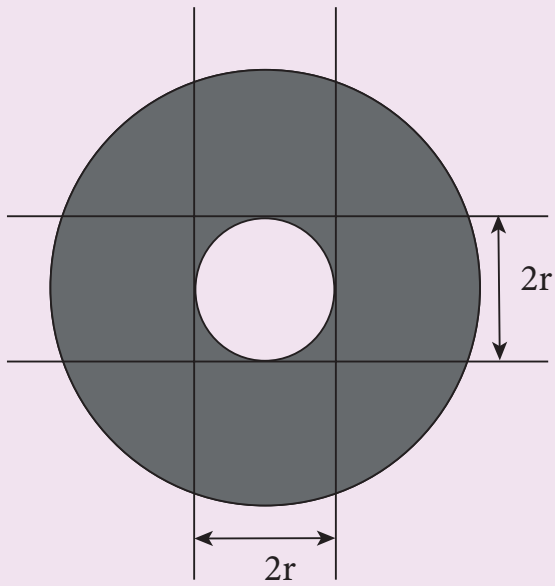
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- Fix a needle near the capillary tube so that the needle touches the water surface
- A Vernier microscope is focused at the water meniscus level and the corresponding reading is taken after making the cross wire coincidence.
- Vernier microscope is focused to the tip of the needle and again reading is taken and noted.
- The difference between the two readings of the vertical scale gives the height (h) of the liquid raised in the tube.
- Now to find the radius of the tube, lower the height of the support base and remove the beaker, carefully rotate the capillary tube so that the immersed lower end face towards you.
- Focus the tube using Vernier microscope to clearly see the inner walls of the tube.
- Let the vertical cross wire coincide with the left side inner walls of the tube. Note down the reading (L_1)
- Turn the microscope screws in horizontal direction to view the right side inner wall of the tube. Note the reading (R_1). Thus the radius of the tube can be calculated as $\frac{1}{2}(L_1 - R_1)$.
- Finally calculate the surface tension using the given formula.



RADIUS OF THE CAPILLARY TUBE

OBSERVATIONS

To measure height of the liquid (h)

Least count of the microscope = _____ $\times 0.001cm$